

Which Imagined Concepts Can Be Steered in LLMs?

Anonymous ACL submission

Abstract

Steering vectors are widely used to control and interpret LLMs, yet benchmarks show that some concepts steer reliably while others barely respond — a variance that is reported but rarely explained. In this research we investigate which kinds of concepts are steerable in Llama-3.1-8B and Gemma-2-9B. We use the RFM method to extract steering vectors (or directions) for approximately 600 concepts drawn from 24 WordNet supersense categories, and validate which of them successfully steer the model. We then build a regression model explaining steerability from concept category, concreteness, polysemy, and word frequency, as well as the share of the top eigenvector during extraction (λ_1 /trace). The results help distinguish concepts that behave as single directions — “primitive” for the model — from those that are composite or context-dependent, and offer practitioners an extraction-time indication of when single-vector steering can be trusted.

- TODO: the prose paragraph above pre-dates the final results — update: add age-of-acquisition to the predictor list, replace “ λ_1 /trace” with “spectrum concentration (participation ratio)”, and consider adding one sentence each on the depth-window finding and the cross-model replication.
- Headline: 41.5% of the 600 concepts steer on Llama (249/600, V1UV6) and 37.3% on Gemma (224/600) under the v4 instrument, and the psycholinguistic laws replicate across BOTH models with all four predictors significant: frequency +, concreteness –, age-of-acquisition –, polysemy –. Spectrum concentration (PR) predicts steerability within both models, at model-specific depths. Frequent, early-acquired, monosemous, abstract concepts steer best.

- Candidate sentence (failure mode): when steering fails, it fails informatively — delivery of the category/frame blend, a diffuse discriminative spectrum, or injection outside the concept’s depth window; not silence. (1 sentence)
- Candidate sentence (instrument): measured steerability is instrument-relative — one changed frame sentence doubles the rate on both models — but the laws survive instrument and model change. (1 sentence)
- Candidate sentence (depth): concepts are injectable in depth windows ordered by abstractness; one well-chosen layer nearly matches all-layer steering. (1 sentence)

1 Introduction (1.0p)

- Hook: steering/RePE papers throw heterogeneous concept lists at models; some concepts steer at 90%, others at chance. The variance is treated as noise; we argue it is signal. (1 para)
- The implicit assumption of top-1 eigenvector / PCA-1 / difference-of-means steering: the concept is representable as one direction, injectable with one coefficient, at whatever layer(s) the method uses. (half para)
- Existing evidence that not all features are one-dimensional (Engels et al.) — but no one has asked which concepts behave as single directions, or whether this aligns with human semantic structure. (half para)
- Our question: is steerability systematic and predictable from the linguistic properties of the word — and what does a steering failure actually consist of? (half para)
- Contributions (short lead-in + three numbered items, 2–3 sentences each): (1 para)

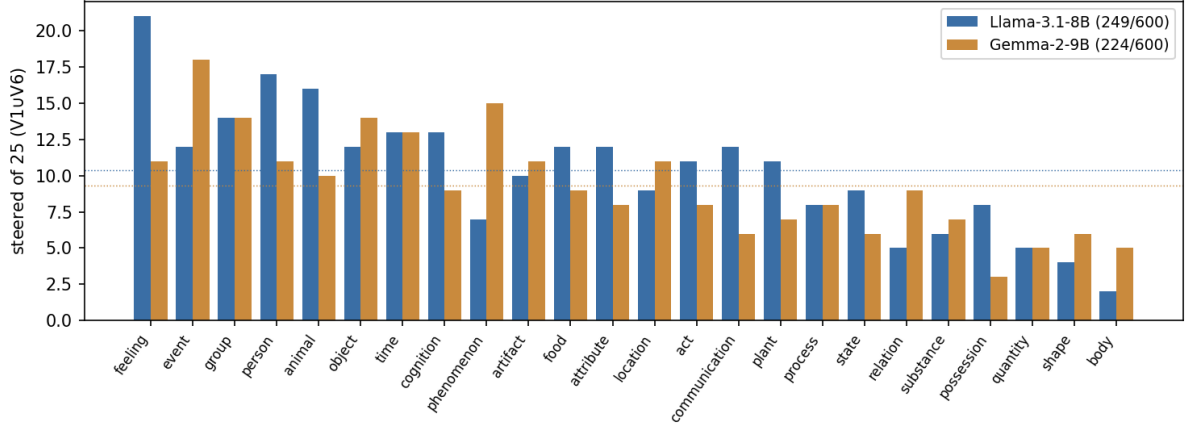


Figure 1: Concepts steered per WordNet supersense (of 25 each; steered = judged as expressing the concept under either evaluation prompt, V1UV6; all-layer steering, v4 extraction frame): steerability varies systematically by category, and the rates and much of the ordering replicate across the two model families.

- C1: a systematic steerability study — ~600 lexicalized concepts across 24 WordNet supersenses, annotated with psycholinguistic norms, steered under a controlled protocol on two models — together with a 7-frame instrument study (Table 2) showing that the extraction contrast itself sets measured steerability (2× rate swing at fixed method and model); per-concept results and code released.
- C2: a regression explaining steerability from frequency, concreteness, age-of-acquisition, and polysemy (all significant, signs replicating across Llama and Gemma), with extraction-internal spectral concentration (participation ratio / λ_1 -share) as a zero-cost predictor that beats the lexical block.
- C3: a mechanistic account of what failure looks like — parent-dominance of extracted directions, coherence $\neq$ identity, injection outside the concept’s depth window — probed causally (within-category subtraction; single-layer depth sweep).

2 Background and Related Work (1.0p)

- Linear representation hypothesis and steering. Park et al. (2024); RepE (Zou et al., 2023); CAA (Panickssery et al., 2024); ActAdd (Turner et al., 2023); ITI (Li et al.,

2023); the RFM benchmark we build on (Beaglehole et al., 2025): kernel ridge + AGOP, top eigenvector as steering vector. Ours: we adopt RFM unchanged and ask which concepts it works for. Second relation: steering rates are instrument-relative — one changed contrast sentence doubles the rate (19.8%→41.5%) at fixed method, model, and judge; method comparisons with hand-tuned per-class prompts inherit this confound. (1 para)

- Multi-dimensional features. Engels et al. (2025): not all features are one-dimensional. Ours: we test the premise on a stratified inventory and tie it to word properties: spectrum concentration (PR / λ_1 -share) predicts steering in both models — “effectively one-dimensional” is measurable and graded, and polysemy acts on steerability largely through it. (half para)
- Hierarchical concept geometry. Park et al. (2025): ~900 WordNet concepts, Llama-3 and Gemma; hierarchically related concepts occupy orthogonal subspaces — the representational-geometry twin of this study. Ours: they characterize how hierarchy is encoded; we test what it predicts about intervention. Our parent-dominance finding is the steering-side counterpart of their parent/child decomposition; and our R4 probe reads their orthogonality causally

149	— the child-specific residual exists but is	199
150	rarely independently injectable (9 gained /	200
151	14 lost of 150). (half para)	201
152	• Predicting steerability. Directional agree-	202
153	ment and geometric predictors of steering	203
154	(un)reliability (Braun et al., 2025, 2026) —	204
155	the closest neighbors to our R3; AxBench	205
156	(Wu et al., 2025) for large-scale evalua-	
157	tion. Ours: we predict from the concept’s	
158	linguistic structure; the lexical laws (fre-	
159	quency +, concreteness −, AoA −, poly-	
160	semy −) replicate across two models and	
161	two instruments — portable predictors en-	
162	gineering signals cannot supply. We add a	
163	caution: geometric predictors are only in-	
164	terpretable relative to the contrast design	
165	(λ_1 -share flipped null→significant with the	
166	frame). AxBench positioning: they sam-	
167	ple concepts from the model (SAE-latent	
168	labels; no norms, no stratification) and	
169	vary methods; we sample from the lexicon	
170	and fix the method — only the latter can	
171	ask which kinds of concepts steer. Their	
172	trained ReFT-r1 bounds our claims: “un-	
173	steerable” means extraction-based rank-1	
174	steering fails. TODO cite: LAP / logit-lens	
175	accessibility (Billa 2026); “When is your	
176	LLM steerable?” (bib stubs). (1 para)	
177	• Causal identification of concept vectors.	
178	Function vectors (Todd et al., 2024); re-	
179	lation decoding (Hernandez et al., 2024);	
180	ROME (Meng et al., 2022). Ours: we iden-	
181	tify by contrast design, probe by interven-	
182	tion (R4). The depth sweep converges with	
183	causal tracing: entity content is injectable	
184	in a sharp mid-stack window (depths 14–17	
185	of 32), where tracing localizes subject en-	
186	richment — but read-side statistics cannot	
187	substitute for write-side tests. (half para)	
188	• Frequency and linear representations. Co-	
189	occurrence thresholds; long-tail knowledge	
190	(Kandpal et al., 2023). Ours: frequency is	
191	a robust positive predictor in both models	
192	— against our pilot expectation, with co-	
193	occurrence-threshold accounts — and thins	
194	when spectrum concentration enters: expo-	
195	sure appears to work partly through repre-	
196	sentational consolidation. (half para)	
197	• LLM–human alignment and psycholinguis-	
198	tic norms. Shani et al. (2025) compare LLM	
	and human concepts on input-side embed-	199
	dings. Ours: mid-stream and causal —	200
	and human structure does predict what can	201
	be steered: early-acquired, frequent words	202
	steer best (AoA $\beta = -0.33 / -0.49$ across	203
	models), over and above frequency. (half	204
	para)	205
	• Methodological caution on cosine. You	206
	(2025): cosine misleads in anisotropic	207
	spaces. Ours: geometry claims rest on dif-	208
	ferential cosines under identical measure-	209
	ment, with the between-class value (0.35)	210
	as the anisotropy control. (half para)	211
	• Gap statement: no prior work connects	212
	concept-level semantic structure to steering	213
	outcome or explains what failure consists of.	214
	Our answer: (i) frequency, AoA, polysemy,	215
	and concreteness jointly predict steerabil-	216
	ity, replicating across Llama and Gemma;	217
	(ii) failure is usually informative — cate-	218
	gory/frame delivery, a diffuse spectrum, or	219
	injection outside the depth window — not	220
	silence. (half para)	221
	3 Defining and Sampling Concepts (1.0p)	222
	• Operational definition. Sidestep the philo-	223
	sophical debate (Murphy 2002; Margolis &	224
	Laurence 1999): a lexicalized concept =	225
	a meaning a speech community assigned	226
	a word to, operationalized as a WordNet	227
	synset; each word mapped to its dominant	228
	(first) noun sense. (half para)	229
	• Inventory. All single-word, noun-dominant	230
	lemmas from WordNet; category = super-	231
	sense (26 lexicographer files; Miller 1990;	232
	Fellbaum 1998; “supersense”: Ciaramita	233
	& Johnson 2003). Exclusions: noun.Tops;	234
	POS-ambiguous lemmas; small cells sam-	235
	pled exhaustively and flagged. (half para)	236
	• Sample (≈ 600 , flat). 25/supersense across	237
	~ 24 usable supersenses; within each su-	238
	persense, stratify over frequency so cate-	239
	gory and frequency are not confounded.	240
	Six supersenses are pre-specified (before ex-	241
	traction) as the featured cells for within-	242
	category analyses: location, person, feeling	243
	(pilot continuity) and artifact, cognition,	244
	food (concreteness extremes). At $n=25$	245
	every cell is adequately powered for class	246

means and absorbs attrition (prompted-ceiling/judge exclusions). (1 para)

- Predictors (one table): Zipf frequency (Speer et al., 2018); concreteness (Brybaert et al., 2014); age of acquisition (Kuperman et al., 2012); polysemy (#WordNet senses); supersense; log category size (#noun synsets in the lexicographer file, covariate). Tested and null: token count, verb-sense count. (half para + table)
- Contrast design. Positives wrap a statement in a universal frame; the frame must be felicitous across all supersenses, else frame felicity confounds category; negatives are the bare statements (benchmark-comparable). Rationale for care here: a contrast in which frame, category, and instance always co-occur cannot statistically separate them (see Pilot) — whatever the frame adds constantly ends up inside every direction. A second negative design — same frame with a different same-supersense concept, which cancels frame and category parent — is used as the causal probe (R4), not the main contrast. (1 para)
- [NEW jul10] Main-run frame: v4 = “Imagine/assume: {c}.” (constant prefix, concept unquoted; negatives bare). Selected from a 7-frame grid on the pilot (Table 2, Appendix B): 77/90 V1UV6 vs. 61 (orig class-specific) and 59 (v2b dual frame) — the only frame at ceiling on moods and far ahead on places. Ablations attribute the gain to assume (bare-noun reading $\approx$ role-assignment, a class-neutral persona prime) and show quoting the concept costs ~ 7 points (mention-mode suppresses enactment). Trade-off, quantified in Appendix B: a constant prefix re-saturates extraction R^2 (the prefix is a free class separator), so the dual-frame diagnostics (informative val- r) are given up in exchange for steering power; the frame-leak caveat transfers to the imagine/assume register (acknowledged in Limitations).
- Pilot/confirmatory split. Pilot = 90 concepts (30 each: places, moods, personalities) from the released benchmark lists, used for all exploration; the WordNet sample is confirmatory, untouched during ex-

ploration. (half para)

4 Method (1.0p)

- Direction extraction (RFM) — and why RFM. Per layer: kernel ridge regression on concept-vs-negative prompts; AGOP $M = \mathbb{E}[\nabla f \nabla f^\top]$; steering vector = top eigenvector of M . Chosen among RepE methods because (a) it is the method of the benchmark we build on (comparability); (b) unlike difference-of-means or PCA-1 it returns the full discriminative spectrum with eigenvalues — the spectral metrics below come for free, and the method is self-diagnosing (the pilot’s confound was discovered through them); (c) its steering performance matches or beats the simpler extractors on the released benchmark. (1 para)
- Spectral metrics. $\lambda_1/\text{tr}(M)$, λ_1/λ_2 , and the participation ratio $\text{tr}(M)^2/\|M\|_F^2$; per-concept score read at a fixed relative depth (per-layer profile retained; the informative depth is model-specific — 40% on Llama, 60% on Gemma). These measure the shape of the discriminative spectrum, not concept identity (see Pilot); they are near-deterministic functions of one another ($|r| \geq .94$), so they enter analyses one at a time. (half para)
- Fixed hyperparameters — a methodological requirement. Kernel bandwidth inflates λ_1/trace mechanically (large bandwidth $\Rightarrow$ near-linear fit $\Rightarrow$ rank-1 AGOP); grid search over near-tied validation correlations injects artifact variance into the dependent measure. All extractions at bw=1, norm=True, reg= 10^{-3} ; vector and metrics read at a fixed (converged) iteration. (half para)
- Two-config steering protocol. Each concept steered under (i) all layers, benchmark defaults (unit vectors, released coefficient grid — comparability); (ii) one mid layer at $\approx 40\%$ depth (vector scaled by the concept’s per-layer projection gap, cf. CAA / ITI; effective single-site dose $\approx 1 \times$ local residual-stream norm). $\Delta(\text{all} - \text{single})$ measures how distributed the concept’s steerable content is — with the caveat, added after the

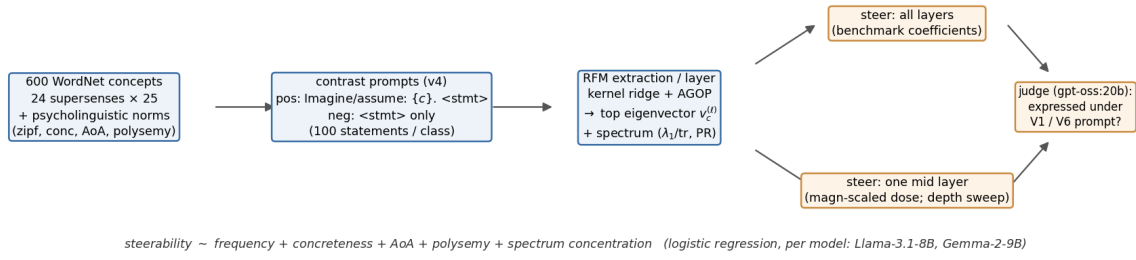


Figure 2: Study pipeline. Each of the 600 WordNet concepts gets a contrast set (v4 frame positives vs. bare-statement negatives), per-layer RFM extraction (top AGOP eigenvector = steering vector; spectrum retained), steering under two configurations, and an LLM-judge evaluation under a cued (V1) and a neutral (V6) prompt; steerability is then regressed on lexical norms and spectrum shape.

depth sweep, that Δ partly measures window placement: the confirmatory single-site arm should sit at depth 15, inside the entity window. Two evaluation prompts (category-inviting V1; neutral V6; listed verbatim); greedy decoding. (1 para)

- Outcome. Primary: steered under V1 or V6 at any coefficient (binary; the union de-biases the evaluation-relativity of either prompt alone, V1–V6 $\phi=0.59$). The binary criterion flattens near-misses (pilot: Edinburgh $\rightarrow$ “Scottish Highlands” scores 0); a graded rubric (V8) re-judges the same generations as a robustness check (Appendix E). Verbatim prompts in Appendix G. (1 para)
- Judge. gpt-oss:20b (local, on the experiment GPU); symmetric judge question across supersenses. Validated against Claude Sonnet 5 on a 360-item set: 87–91% agreement, $\kappa = .72-.78$; a stricter candidate (Haiku 4.5) was rejected for one-directional divergence from the two-judge consensus. Confirmatory-study additions: re-validation on ~ 200 WordNet-distribution items + human spot-check of disagreements (Appendix F). (half para)
- Controls: prompted ceiling. One unsteered generation per concept (“Tell me about ‘{c}’”) + judge: 578/600 (Llama) and 586/600 (Gemma) concepts are expressible when simply asked, and the residual “failures” are largely judge strictness on polysemous items (ray, seven, junk); several of them nonetheless steer. Ceiling-

conditional steering rates are indistinguishable from unconditional (Llama 241/578 = 41.7% vs. 41.5%). Decisively for the bottom cells: every noun.body concept is expressible (25/25) — its 2/25 steering rate is a genuine steering floor, not an expression floor. Pipeline additionally sanity-checked on known-good concepts per model. (half para)

- Models. Full protocol on Llama-3.1-8B-Instruct; headline analyses replicated on Gemma-2-9B-it. (half para)

5 Pilot Diagnostics and Protocol Selection (1.0p)

- Hyper-parameter selection (summary). Validation r saturates ($\sim 0.97-0.99$) across bandwidths, statement counts, and iterations $\Rightarrow$ selection by validation is third-decimal noise; large bandwidth forces rank-1 AGOP by construction; converged readout cuts metric instability 3–4 $\times$. Decision rule: fixed modal setting, converged readout. Details in Appendix A. (NB the saturation itself is frame-structural: any constant-prefix contrast hands the classifier a free separator; dual frames de-saturate val- r — Appendix B.) (half para)
- Finding 1: parent-dominance. Different same-category concepts’ extracted top-1 directions are nearly identical (places: cosine $0.97 \rightarrow 0.84 \rightarrow 0.72$ across depth; all top-3 components ≥ 0.90 shared). Single-layer dose-response saturates at the category prototype (every city $\rightarrow$ “quiet

beach at sunset”) with region-level near-misses (Edinburgh→Dublin pub); giving secondary components injection mass adds a generic roleplay axis, not instance content. Three legs: geometry, dose-response, component ablation. Interpretation: a contrast in which frame, parent, and instance always co-occur is statistically one axis — what steering delivers is mostly the category. (1 para + figure parent_share_by_depth.png)

- Finding 2: coherence $\neq$ identity. Places show the highest band λ_1 /trace yet fail single-layer steering; moods (lower) succeed. Spectral metrics certify that a coherent (context-invariant, rank-1) direction was extracted — not that it is the judged content. Consequence: their regression coefficients are expected to be conditional on contrast type; tested under both contrasts. (1 para)
- Finding 3: attribute vs. instance. Attribute concepts (moods) steer from one mid layer; instance identities steer only under all-layer injection, reaching region granularity — consistent with multi-stage assembly (per-layer residue accumulation + late lexical tipping + autoregressive bootstrap once the name is emitted). Motivates the two-config protocol and Δ . (1 para)
- Protocol justifications (one line each; evidence in Appendix C): single mid layer in the mid-stack behavioral window — the depth sweep (Discussion) later located it precisely at depths 14–17 for entity concepts, with the original 40% (depth-13) site just below its onset; 60% yields token-bias then repetition collapse. No top- k arms — the top-3 subspace under the released contrast is category-shared. Coefficient $\approx 1 \times$ local residual norm via per-layer magn scaling. Granularity as secondary outcome — the binary judge flattens near-misses. (1 para)

6 Results (1.5p)

- R1: Steerability varies systematically — 249/600 (41.5%) steer (Llama, all-layer, v4 frame, judge = gpt-oss; V1 191, V6 195, union V1UV6; Figure 1). Per-supersense

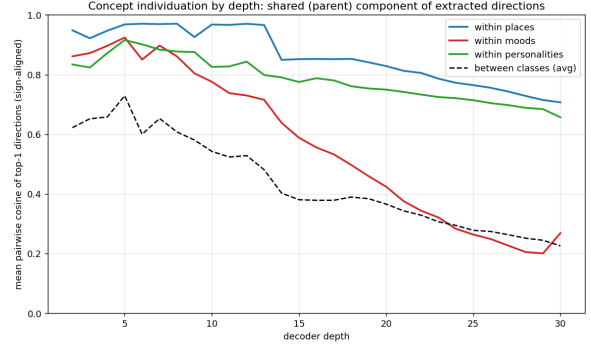


Figure 3: Pilot Finding 1 (parent-dominance). Cosine similarity of same-category concepts’ extracted top-1 directions across depth: the released contrast yields largely shared (category) directions for places, while moods individuate.

spread 2–21 of 25: top noun.feeling (21), noun.person (17), noun.animal (16); bottom noun.body (2), noun.shape (4), noun.quantity/noun.relation (5). (1 para + figure)

- R1 instrument comparison (v2b arm): the frame moves the rate, much less the ordering. Under the dual v2b frame the same protocol yields 119/600 (19.8%; spread 2–9); switching to v4 improves 22 of 24 supersenses while the ranking stays broadly stable (feeling top, body bottom under both; noun.body immovable at 2/25; Figure 5, Appendix B). V1–V6 agreement $\phi = 0.59$ — measured steerability is partly evaluation-relative, motivating the union outcome. (half para)
- R2: Psycholinguistic predictors (Table 1): all four lexical predictors significant, same signs in both models. Llama: conc -0.20^* , zipf $+0.28^*$, AoA -0.33^{**} , polysemy -0.30^{**} ; Gemma: conc -0.20^* , zipf $+0.23^\dagger$, AoA -0.49^{**} , polysemy -0.35^{**} . The pilot expectation (concreteness +, polysemy carrying the signal, frequency null) is rejected: frequent-but-abstract concepts steer best. AoA and polysemy were null under v2b and surface once the instrument improves (v2b: zipf $+0.40^*$, conc -0.28^* ; Appendix B). Honest-reporting nuance (Figure 4): polysemy and concreteness do not separate univariately (MW $p > .6$; effects conditional on correlated covariates); zipf, AoA, PR separate cleanly in both mod-

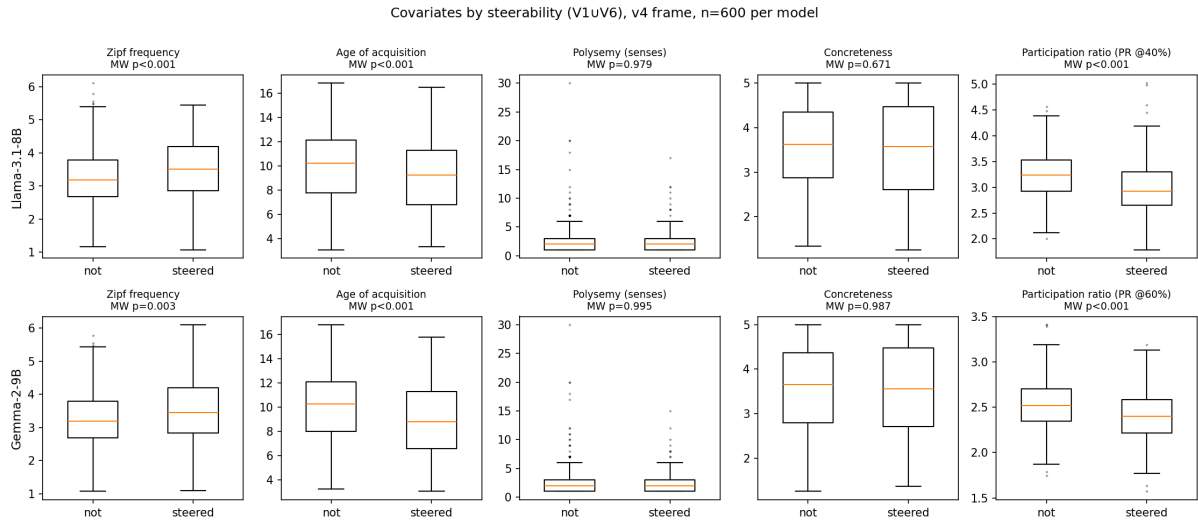


Figure 4: [NEW jul10] R2/R3: covariates by steering outcome (v4 runs, $n=600$ per model; PR at each model’s informative depth). Zipf, AoA, and PR separate univariately in both models; polysemy and concreteness do not (their regression effects are conditional on correlated covariates).

els ($p < .005$). Collinearity among lexical predictors is benign (conc–zipf $r = .09$; VIF < 2.7). Log category size is a positive predictor in Llama (+0.31** with the lexical block; +0.21* surviving PR) but null in Gemma (−0.01) — reported as a non-replicating candidate, not among the laws. (1 para + table)

- R3: Spectral metrics: every spectral metric carries signal, and each alone beats the whole lexical block. Llama, one-at-a-time: λ_1/trace +0.55** (AIC 778.5), λ_1/λ_2 +0.57** (775.3), PR −0.49** (786.5) vs. lexical block 799.0, null 816.4; best model lex+PR (763.6, PR −0.62**). Concentrated discriminative spectra steer; polysemy is largely absorbed by PR in the joint model — polysemous words appear to fail via diffuse spectra. Instrument caution (one of our sharpest methodological findings): under the dual v2b frame λ_1/trace was null — the register axis parked in within-positive variance can capture the top eigenvector for weak concepts (Appendix B), corrupting spectral readouts. Spectral diagnostics are only interpretable relative to the contrast design. Spectral metrics are near-collinear ($r \leq -.94$): one-at-a-time only. (1 para)
- Category is partly mediated by measurable word properties. Supersense dummies beat

the null (AIC 784.1 vs. 816.4) but lose to lexical+PR (763.6), and adding them to the best model does not pay: the category signal visible in R1 is substantially carried by the continuous predictors. (Under v2b the dummies did not even beat the null; the wider v4 spread carries real categorical signal.) (half para)

- R4: The rescue probe (causal). Design: for failing concepts, cancel the frame/parent component — either by re-extracting with within-category contrasts (“loves Edinburgh” vs. “loves {other city}”; not yet run) or by subtracting the leave-one-out supersense mean from the saved direction (v_{spec} ; executed, $n=150$). Outcome: no net rescue (25 vs. 30 of 150 steer; McNemar $p \approx .4$) — shared-component dominance is not the prevailing failure mode, though the subtraction doubles as a frame-leak control. Details moved to Appendix Z3 [bold note: full v_{spec} material relocated there; inclusion in the final submission undecided]. (1 para)
- R5: Distributedness. $\Delta(\text{all-layer} - \text{single-layer})$: single-layer depth-13 (magn dose) 90/600 (15.0%) vs. all-layer 249 (41.5%), ratio 2.8× (v2b: 3.5×); 11 concepts steer only single-layer (over-steer rescues; abstract/attribute skew). Caveat from the depth sweep: depth 13 sits below the en-

	Llama-3.1-8B		Gemma-2-9B	
	lexical	+PR	lexical	+PR ^{60%}
concreteness	-0.20*	-0.43**	-0.20*	-0.18†
Zipf frequency	+0.28*	+0.23	+0.23†	+0.20
AoA	-0.33**	-0.28*	-0.49**	-0.47**
polysemy	-0.30**	-0.15	-0.35**	-0.29*
n tokens	+0.04	-0.05	+0.06	+0.02
participation ratio		-0.62**		-0.39**
AIC	799.0	763.6	767.3	750.9

Table 1: Main regression (v4 instrument). Logistic regression of steerability ($V1 \cup V6$) on z-scored predictors, $n=600$ per model, all-layer config. PR read at each model’s informative depth (Llama 40%, Gemma 60% — Gemma’s spectral signal is null at the 40% convention and switches on at 60–70%). Reference AICs — Llama: null 816.4, supersense-only 784.1, one-spectral-alone 775–787; Gemma: null 794.8. Spectral collinearity is extreme under v4 ($PR-\lambda_1/\text{trace } r \leq -.94$ both models): spectral metrics are reported one-at-a-time only. † $p<.1$, * $p<.05$, ** $p<.01$.

tity window, so Δ partly measures window placement — interpret jointly with the sweep (Discussion). (half para)

- R6: Generality (Gemma-2-9B replication): rates and laws replicate; item convergence is itself instrument-dependent. Rates: 224/600 (37.3%) vs. Llama 249 (41.5%); the v4 gain replicates ($\sim 2\times$ both models’ v2b rates). Laws: all four lexical signs replicate (AoA -0.49^{**} strongest); PR replicates at Gemma’s informative depth (60–70%: -0.42^{**} ; null at Llama’s 40% convention) — same law, model-specific depth. Items: per-concept $\phi = 0.43$ (154 shared) and supersense-profile $r = 0.44$, both up from v2b (0.33 / 0.12 n.s.): part of the earlier item-divergence was instrument noise, though which concepts steer remains substantially model-relative (Gemma tops: event/phenomenon/group; body bottom in both). Supporting pilot evidence: the λ_1/trace class ordering inverts across models (Llama places > moods; Gemma the reverse) — spectral values are model-geometry-laden; compare within-model only. Gemma dose grid [14,17,20,23] set by dose-response audit (Appendix C). (1 para)

7 Discussion (1.0p)

What makes a concept steerable: attributes (context-general colorings) admit a single injectable direction; instance identities require assembled, multi-stage content — steering failure is usually delivery of the parent category, not silence. (1 para)

What the spectral metrics are: an intervention-match score, with an instrument caveat. Spectrum concentration (low PR / high λ_1 -share) measures how well the concept’s discriminative signature fits the one move single-vector steering can make (one constant vector, all contexts); it predicts steering in both models. The mechanism behind R2’s reversal: abstract (not concrete) concepts steer better — for concrete instance-like concepts the dominant direction is largely the shared category axis, so its coherence does not deliver the instance (parent-dominance). And the caveat: spectral diagnostics of extracted directions are only interpretable relative to the contrast design that produced them — under the dual v2b frame λ_1/trace read as null because a frame-register axis could capture the top eigenvalue slot (Appendix B); under the constant v4 prefix it predicts steering with the expected sign. (1 para)

- Measured steerability is instrument-dependent — and so is cross-model generality. Changing one frame sentence doubles the steerable share in both models (19.8→41.5% Llama; 18.8→37.3% Gemma) and raises cross-model item agreement (ϕ 0.33→0.43) and category-profile agreement (r 0.12→0.44). Absolute rates from any single protocol should not be read as properties of the model alone; but the laws (frequency +, concreteness –, AoA –, polysemy –, spectrum concentration –) survive instrument change and model change, which is the paper’s strongest generality claim. (half para)
- Depth structure: concepts live in depth windows, ordered by abstractness. Attributes are injectable from $\sim 30\%$ depth onward with a broad plateau; entity identities switch on sharply at $\sim 45\%$ (window 14–17 of 32) and fade late; personas need multi-site injection. One well-chosen layer nearly matches all-layer steering (71 vs. 77 of 90)

639	— “all layers” mostly buys window cover-	• Result 3: one well-chosen layer $\approx$ all layers	686
640	age. Read-side extraction statistics do not	(71/90 at depth 15 vs. 77/90 all-layer): the	687
641	locate the window (they miss the late de-	all-layer advantage is largely not knowing	688
642	cline entirely): injectability is a write prop-	the window, not distributed injection per se	689
643	erty. (half para)	(except personalities). Per-layer read-side	690
644	• Status of extracted vectors: the resid-	metrics rise into the window but entirely	691
645	ual stream has no privileged basis, so ex-	miss the late decline ($ r \approx .5-.6$): first-	692
646	tracted directions are experiment-relative;	steerable-depth is a write property. (half	693
647	within-model relational claims are valid,	para)	694
648	coordinate-level and raw cross-model com-	• Future work: same sweep on the	695
649	parisons are not. Anchored spaces (em-	six featured WordNet cells (150 con-	696
650	bedding/unembedding, MLP neurons) are	cepts) with taxonomic depth as ground	697
651	where mechanism pins a frame. (half para)	truth (hypernyms shallower than hy-	698
652	• Two routes to a vector’s identity: exper-	ponyms?); independence from the spec-	699
653	iment design (contrasts that vary exactly	tral axis assessed within supersense;	700
654	one thing) vs. causal mediation (function	compositional vignette — is direc-	701
655	vectors, ROME). Pipeline: propose by con-	tion(“strict father”) $\approx \alpha$ direction(“strict”)	702
656	trast, accept by causal test — R4 instanti-	$+\beta$ direction(“father”)? (half para)	703
657	ates it. (half para)		
658	• Practical implications for RePE: report	8 Limitations (0.5p; uncounted in ACL	704
659	spectral concentration and contrast de-	format)	705
660	sign alongside steering benchmarks; strati-	• Single extraction method (RFM); λ_1 /trace	706
661	fied concept inventories; granularity-graded	is AGOP-specific — PCA explained-	707
662	judging over binary. (half para)	variance as cross-check (appendix if time).	708
		(half para)	709
663	The depth axis: first-steerable-depth sweep	• Steering conclusions are relative to the ad-	710
664	(0.5p)	ditive intervention class; clamp/projection	711
665	• Design. Pilot 90	interventions may behave differently (tar-	712
666	(moods/places/personalities) $\times$ every	gets saved, untested). (half para)	713
667	single-layer depth 10–19, magn dose, v4	• LLM judge (validated but imperfect); En-	714
668	directions, V1UV6 (Figure 9). (half para)	glish only; dominant-sense assumption;	715
669	• Result 1: entity concepts have a sharp on-	norms are human ratings. (half para)	716
670	set at depth 14 (of 32). Places: 0–2/30 at	• Correlational core with one causal probe	717
671	depths 10–13, then 22/30 at depth 14, win-	(R4); extraction-design effects and model	718
672	indow 14–17, fading by 19. This dissolves the	limits not fully separable outside that de-	719
673	earlier “places don’t steer single-layer” con-	sign. (half para)	720
674	clusion: the old two-site protocol (13/19)	• Depth-sweep evidence is pilot-only ($n=90$,	721
675	straddled the window. (half para)	3 concept classes, one model); the	722
676	• Result 2: attribute concepts steer shal-	WordNet-wide sweep and the Gemma	723
677	low and everywhere. Moods: 16/30 al-	sweep are future work. Measured rates	724
678	ready at depth 10, plateau 25–30 from	are instrument-relative (quantified, but a	725
679	depth 13 through 19 — attributes become	reader should not quote “41.5%” as a model	726
680	injectable ~ 4 layers before instances (the	constant). (half para)	727
681	abstraction-depth hypothesis, confirmed or-	• The v4 frame’s constant prefix puts an	728
682	dinally). Personalities never find a single-	imagine/assume register component into	729
683	layer home (peak 20, collapse to 1 by depth	every direction; the frame-leak control	730
684	19) — persona content appears genuinely	(v_{spec} subtraction, Appendix Z3) was run	731
685	multi-site. (half para)		

on the v2b instrument only and not repeated for v4 — some v4 successes may be partly frame-carried. (half para)

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A Appendix A: Fixing the RFM hyper-parameters

- + Validation r (probe accuracy) saturates (~ 0.99) across bandwidths, statement counts, and RFM iterations $\Rightarrow$ any selection by $\text{val-}r$ is third-decimal noise; it is also a poor steerability predictor. Boxplots: `box_rsqr.png`.
- + λ_1/trace at bandwidth $L=1$ discriminates across concepts; $L=10$ collapses it toward 1 (rank-1 AGOP by construction) — so the bandwidth choice mechanically sets the spectral metrics; fixing it is a validity requirement, not a convenience. `box_l1trace.png`, `box_l1l2.png`.
- + The released readout (metrics at the best- r iteration) is unstable — it samples a value climbing $0.0002 \rightarrow 0.17 \rightarrow 0.46$ across iterations at a noise-chosen point; reading at convergence cuts the statement-count swing 3–4 $\times$. Both vector and metrics read at a fixed (converged) iteration.
- + Final setting: `bw=1`, `norm=True`, `reg=10-3`, converged readout; saved per-layer stats (top-3 evals, `magn`, `tpos/tneg`, `hnorm`) in the `rfmstats` pickles.
- ? Per-concept “best layer” by `argmax` is unstable: $\text{val-}r$ best is noise-driven; λ_1/trace and λ_1/λ_2 are bimodal with a degenerate spike at the first block (exclude early blocks). `besthist_*.png`.
- ? Sweep factors (probe set of 24 concepts, one per supersense): bandwidth $\times$ normalization $\times$ contrastive construction $\times$ iteration readout; criteria: stability, discrimination, semantic signal, steering non-degradation. Artifacts: `hpsweep_phase2_*`.

B Appendix B: Choosing the extraction frame

- + Frame grid (Table 2; pilot 90, all-layer $K=1$, gpt-oss judge). A universal frame is required for the 24-supersense inventory (no class template exists for arbitrary nouns), and it sets the construct. $V1 \cup V6$ totals: orig 61, v2 61, v2b 59, v3 45, v4 77, v4b 59, v4c 66, v4d 56.

- Frame-vocabulary leak (v2). A dual frame cancels only what differs between its two variants; the shared semantics (fascinated $\cap$ preoccupied = absorption) stays in the class mean and leaks into every direction — steered moods embody distraction, costing moods 30 $\rightarrow$ 22 while lifting places 2 $\rightarrow$ 10.
- Weakening the verbs does not fix it (v3). Light frames give fainter directions, not cleaner ones: under-steering everywhere (41/90), worst of all dual frames.
- + Register split works partially (v2b). Affective $\times$ epistemic verbs shrink the shared core to “strong engagement”: places best-ever among dual frames (14/13), at a cost to moods (22 $\rightarrow$ 17). The frame acts as a register selector.
- + The v4 family: a single constant hypothetical prefix beats everything. v4 = “Imagine/assume: {c}.” reaches 77/90 — at ceiling on moods (29–30) and far ahead on places (16 V1 / 22 V6, vs. 2–4 under orig); personalities are frame-insensitive. Ablations: quoting the concept costs ~ 7 points, concentrated in embodiment classes (v4c 66 vs. v4b 59) — quotes mark metalinguistic mention; dropping “/assume” costs ~ 11 points, almost all places (v4 77 vs. v4c 66); the slash construction has no magic of its own (“Imagine/consider” is worst, 56). Reading: bare-noun “assume: X” resolves as role-assignment — a class-neutral persona prime; this also revises the v3 lesson: low-affect frames can work when the prefix is constant (its content sits in the class mean and supplies drive) rather than dual (it cancels). Caveat: single runs at $n=30/\text{class}$ (± 3 –5 judge noise); 59-vs-56 are ties, 77-vs-66 is structure. v4 promoted to the main-run instrument (Llama 249/600, Gemma 224/600; RUNLOG 019/022); v2b retained as the instrument-comparison arm (Figure 5).
- + What the frame does to extraction R^2 . Constant-prefix frames (orig, v4 family) saturate $\text{val-}R^2$ at .985–.998 for every concept — the prefix is a free class separator (Fig. 8); dual frames de-saturate it into a per-concept signal-quality measure (v2b

937	median .716, q10 .30) that predicts steering	tary slices while extra components at one	985
938	(MW $p<.001$). One diagnostic axis is given	layer are not (see Appendix D).	986
939	up for steering power.		
940	+ Why dual frames depress val- r — the	+ Depth sweep (pilot 90, v4 directions,	987
941	axis-alignment diagnostic. Same 90 con-	single-layer magn, depths 10–19; Fig. 9).	988
942	cepts, same code: orig .997 vs. v2b .716	Totals 31/33/23/38/64/71/54/61/42/34	989
943	(paired $\Delta=.34$); an old-vs-current-code re-	(d10→d19; V1∪V6). Places: 0–2 through	990
944	run is bit-equivalent ($\cos 1.0000$), exonera-	d13, 22/30 at d14, window 14–17 — the	991
945	ting the pipeline. Recomputing activations	earlier two-site protocol (13/19) straddled	992
946	for the 4 lowest- and 4 highest- R^2 WordNet	it, producing the false “places don’t steer	993
947	concepts: high- R^2 top-1 directions align	single-layer.” Moods: onset ≤ 10 , plateau	994
948	with the label axis ($ \cos .65-.87$), low-	25–30 through d19. Personalities: no	995
949	R^2 ones with the fascinated-vs-expert frame	single-layer home (peak 20, collapse to 1	996
950	axis (.81–.93) — the deliberately-inserted	at d19). Peak single-site d15 = 71/90 vs.	997
951	within-positive frame variance wins the top	all-layer 77. Per-layer read metrics rise	998
952	eigenvalue slot when the concept is weak,	into the window but stay flat through the	999
953	so $K=1$ injects a register axis. The raw	late decline ($ r \approx.5-.6$): the window is not	1000
954	class-mean difference is healthy for all eight	recoverable read-side. RUNLOG 021.	1001
955	(norms 5.5–7.3): the failure is eigenvalue	– Single-site depth-19 (pre-sweep evidence,	1002
956	ranking, not absent signal; entangled cases	orig frames): no behavioral window — lex-	1003
957	(princess: $ \cos (\text{concept, frame}) = .37$) are	ical word-lists (bitter → “bitter, cold, dry,	1004
958	register-colored concepts.	sarcastic”) → repetition collapse within	1005
959	+ At the supersense level, mean extraction fit	0.5–2× local stream norm. Late injection	1006
960	tracks steering in Llama ($r=.67$; Fig. 6) but	biases output tokens, not behavior.	1007
961	not Gemma (all-layer median readout; cf.	+ Single-site depth-13 (pre-sweep, orig	1008
962	the model-specific informative depth).	frames): wide behavioral window but	1009
963	+ Consequence for spectral predictors:	places 0/15, saturating at the category	1010
964	contrast-dependence. Under v2b, λ_1/trace	prototype (“quiet beach at sunset” for	1011
965	is null and sign-flipped as a steerability	every city) with region-level near-misses	1012
966	predictor (its pilot inversion — places	(Edinburgh→Dublin pub). Coefficient is	1013
967	highest λ_1/trace yet failing single-layer	not the limiting factor; vector content is.	1014
968	— is the same lesson: the metric certifies	+ All-layer steering (benchmark de-	1015
969	coherence of the extracted blend, not	faults): places reach region granularity	1016
970	its identity); under v4 it is significant	(Edinburgh→Scottish Highlands with Scots	1017
971	alone (+0.55**) with the pilot’s sign (pilot	dialect) — per-layer residue accumulation	1018
972	depth-9 CV-AUC ≈ 0.86 , Fig. 7). v2b-	+ late lexical tipping + autoregressive	1019
973	instrument regressions for comparison: zipf	bootstrap.	1020
974	+0.40*, conc -0.28^* , AoA/polysemy null,	+ Single-layer arm of the v4 600-run (depth	1021
975	PR -0.27^* ; robustness checks that car-	13, magn): 90/600 (15.0%) vs. all-	1022
976	ried over to v4: lexical collinearity benign	layer 249; 11 single-layer-only rescues (ab-	1023
977	(conc–zipf $r=.09$), mixed-effects reproduces	stract/attribute skew). NB depth 13 sits	1024
978	signs, spectral metrics one-at-a-time only	below the entity window found by the sweep	1025
979	($ r \geq .85$).	— the confirmatory single-site arm should	1026
		move to depth 15. RUNLOG 020.	1027
980	C Appendix C: Steering layers and	? Sample note: the pre-sweep depth bullets	1028
981	coefficients	report interim $n=15/\text{class}$ shakedowns un-	1029
982	+ Cross-layer directions are near-orthogonal	der orig frames; the sweep re-measures at	1030
983	(cosine ≈ 0.07): the concept is re-encoded	$n=90$ and supersedes the two-site depth	1031
984	per layer, so multiple layers are complemen-	conclusions. Qualitative patterns (proto-	1032
		type delivery, token-bias late) held.	1033

frame	class	V1	V4	V6	V1UV6	med R^2
orig (class-specific)	pers.	24	4	17	26	.996
	moods	30	27	27	30	.996
	places	2	14	4	5	.998
v2 “fascinated / preoccupied”, quoted	pers.	24	3	9	24	.804
	moods	22	16	24	26	.740
	places	10	4	3	11	.807
v2b “fascinated / expert on”, quoted	pers.	22	–	13	23	.715
	moods	17	–	14	19	.785
	places	14	–	13	17	.531
v3 “thinking about / on your mind”, quoted	pers.	18	0	8	20	–
	moods	16	13	9	17	–
	places	7	9	5	8	–
v4 “Imagine/assume: {c}.”	pers.	23	9	9	23	.997
	moods	29	28	27	30	.995
	places	16	17	22	24	.992
v4b “Imagine: ‘{c}.’” (quoted)	pers.	20	2	10	23	.997
	moods	19	20	14	22	.986
	places	10	13	12	14	.990
v4c “Imagine: {c}.”	pers.	23	4	10	25	.997
	moods	28	27	24	28	.993
	places	8	12	12	13	.994
v4d “Imagine/consider: {c}.”	pers.	21	3	5	21	.996
	moods	21	23	17	22	.985
	places	10	10	13	13	.991

Table 2: Extraction-frame grid (pilot $N=90$: 30 personalities / 30 moods / 30 places; Llama-3.1-8B, all-layer $K=1$, default coefficients, gpt-oss judge). Cells = concepts steered out of 30 under each evaluation version; V1UV6 = union outcome; med R^2 = class median of per-concept median validation R^2 across layers. Constant-prefix frames (orig, v4 family) saturate R^2 by construction (the prefix is a free class separator); dual frames (v2, v2b) make R^2 concept-informative; v3 direction files were not retained, so its R^2 cells are unavailable. V1UV6 totals: orig 61, v2 61, v2b 59, v3 45, v4 77, v4b 59, v4c 66, v4d 56.

1034	– Two mid layers (9+17) was the best	norm (hnorm); magn/hnorm $\approx$ 0.4–0.5	1058
1035	targeted multi-site result ($\sim 48/35$ of 90);	mid-stream $\Rightarrow$ multiplier ≈ 2 –3; magn $\approx$	1059
1036	9+26 worse (late layer contributes generic	norm-relative scaling ($r=0.90$), so the two	1060
1037	content); late-only steering fails outright.	schemes are redundant.	1061
1038	[struck jul10: combo-layer arm judged non-		
1039	essential]	+ Gemma-2-9B cross-model calibration.	1062
		Gemma-2’s residual norms are $\sim 50\times$	1063
1040	– Single-layer steering at depth 9: moods	Llama’s (median hnorm 484 vs. 9.4), yet	1064
1041	steer, experts middling, places ≈ 0 . [struck	the usable all-layer window is 12–21, not	1065
1042	jul10: superseded by the depth sweep]	$50\times$ Llama’s 0.5–0.8: injection compounds	1066
1043	+ Coefficients: fixed/hand-swept values are	over 41 layers and degenerates above ~ 25 .	1067
1044	brittle — the working value shifts $\sim 10\times$	Norm-relative reasoning brackets but does	1068
1045	with depth and with the number of steered	not pin the dose; a short empirical sweep is	1069
1046	layers.	needed per model family.	1070
1047	+ magn (natural projection gap, saved per	+ Gemma dose-grid shift by dose-response au-	1071
1048	layer) grows $\sim$ exponentially with depth	dit. In the v2b-600 run, successes con-	1072
1049	(Fig. 10); scaling each layer’s vector by	centrate monotonically toward the top of	1073
1050	magn makes the coefficient a dimension-	the [12,15,18,21] grid (uniquely at 21 for	1074
1051	less multiplier (cf. CAA magnitude; ITI σ -	12–13 concepts) and degeneration only be-	1075
1052	scaling). $\sqrt{N}$ count-normalization (cross-	gins at 21 — the grid truncated the dose-	1076
1053	layer orthogonality $\Rightarrow$ quadrature) makes	response on its rising side. v4 grid shifted	1077
1054	the multiplier transfer across layer counts;	to [14,17,20,23]; best-over-coefficients scor-	1078
1055	usable band centers near ~ 1 .	ing makes the added top dose risk-free.	1079
1056	+ Single-site calibration: effective single-layer	? Clamp — / — project-to-target (set the	1080
1057	steering needs $\approx 1.0\times$ the local residual	projection to the concept’s natural	1081

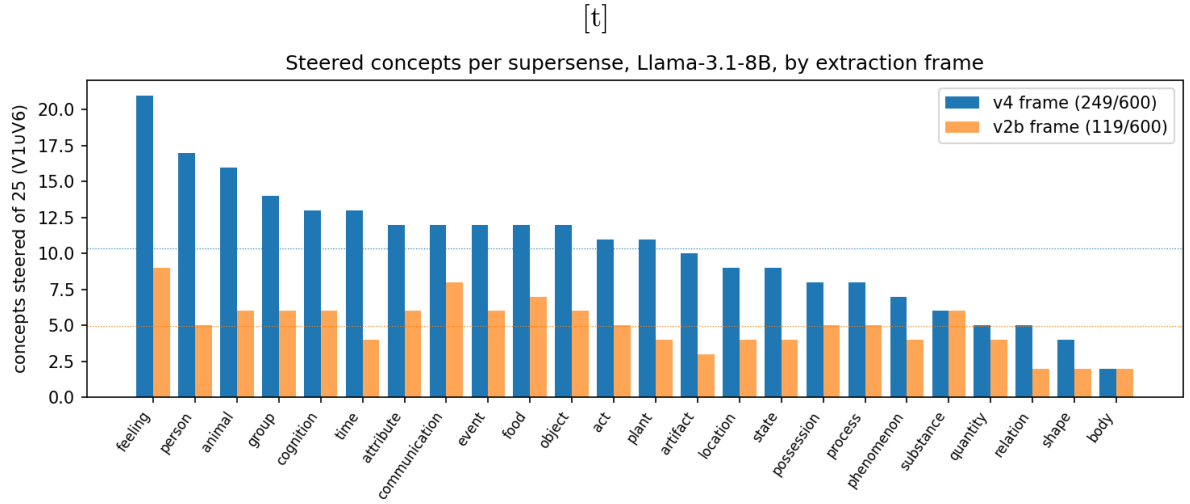


Figure 5: Steered concepts per supersense (of 25; V1UV6; Llama-3.1-8B, all-layer config), under the main v4 instrument and the v2b instrument-comparison arm. The frame moves the rate ($\sim 2\times$), far less the ordering (feeling top, body bottom under both).

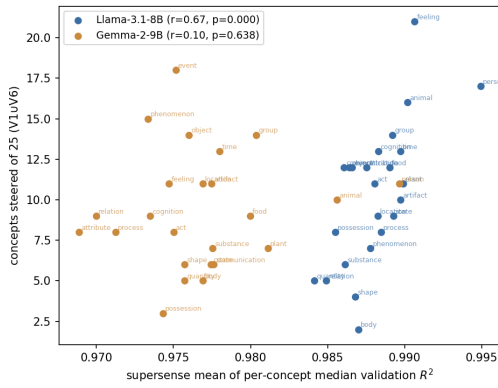


Figure 6: Appendix B: supersense-level extraction fit vs. steering (v4 runs). Per supersense: mean of per-concept median validation R^2 (x) against steered count (y). Llama: $r=.67$, $p<.001$ — with noun.body the “cleanly separable yet not injectable” outlier; Gemma: $r=.10$, n.s. (its informative extraction signal sits deeper than this all-layer median readout).

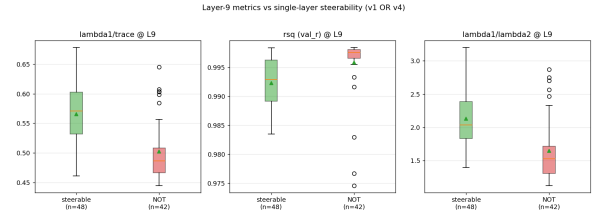


Figure 7: Appendix B: pilot spectral metrics at depth 9 by steering outcome (grouped CV-AUC ≈ 0.86 , orig frames).

D Appendix D: Is more than one direction better? ($K>1$)

- Top- k at one layer, $k=3$: with eigenvalue weights $\approx k=1$; with equal weights worse (moods 15 $\rightarrow$ 10) — components 2-3 are a generic roleplay/self-narration axis (“my name is Luna”), 0.92/0.90 shared across cities.

- ? Parent-dominance bounds what $K>1$ can recover: all top-3 components are ≥ 0.90 shared across same-category concepts at every depth (places cosine 0.97 $\rightarrow$ 0.84 $\rightarrow$ 0.72; moods individuate to 0.46) — under a contrast in which frame, parent, and instance always co-occur, the extra components carry the shared axes, not the missing instance content. plots/parent_share_by_depth.png.

- ? Participation ratio $\approx 2-3$ for all concepts at all layers — the discriminative subspace at

positive-class value; tpos/tneg saved) is the one structurally different alternative — bounded, token-dependent, self-allocating across layers; untested. [struck jul10: not run]

- ? Process lesson: three config-state incidents share one root cause — hand-set knobs without provenance; fix by run tags in artifact filenames. [struck jul10: engineering note, not results]

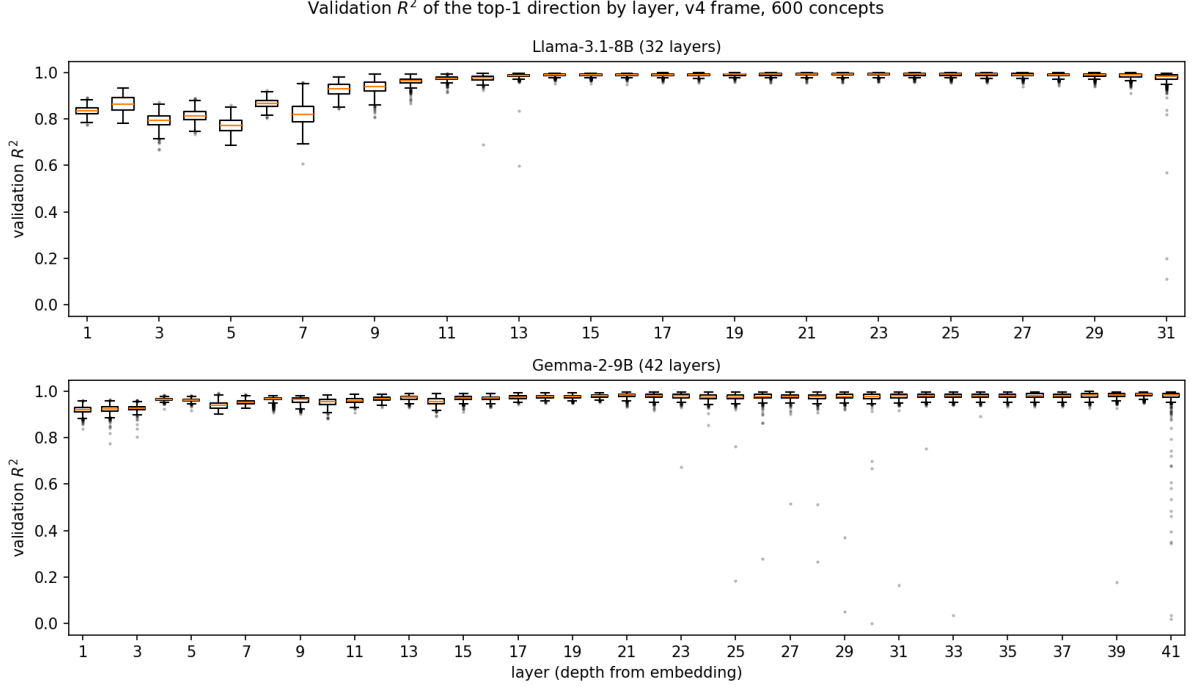


Figure 8: Appendix B: validation R^2 of the top-1 direction by layer under the v4 (constant-prefix) frame, 600 concepts, both models: near-saturation at every usable depth — the constant prefix is a free class separator, so extraction fit carries little per-concept information (contrast with the dual v2b frame, median .716 with a long informative tail; Appendix B bullets).

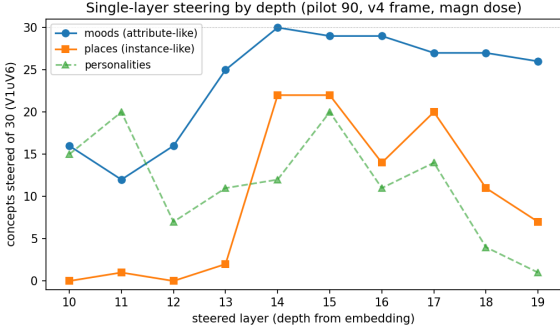


Figure 9: Retention by steered depth (pilot 90, v4 directions, single-layer magn dose). Attributes (moods) steer shallow and everywhere; entity identities (places) switch on sharply at depth 14 (of 32), window 14–17; personas never find a single-layer home.

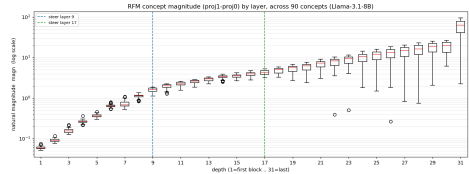


Figure 10: Appendix C: the per-layer natural projection gap (magn) grows ~exponentially with depth, motivating magn-scaled dosing.

1112 one layer is effectively rank 2–3, bounding
1113 top- k from above. PR_depth9_hist.png.

1114 ? Open fix candidate (from the Appendix B
1115 diagnostic): select the saved component by
1116 label correlation instead of eigenvalue — for
1117 frame-captured concepts the concept axis
1118 plausibly sits in component 2–3; untested
1119 at scale.

E Appendix E: Choosing the evaluation prompts

+ Evaluation versions. V1 = category-inviting question (“What is your favorite subject?”) — collapses the response space so a weak direction only tips a forced choice; V6 = neutral open field (“What would you like to talk about today?”) + symmetric judge crediting aboutness or embodiment — only strong topic-steering surfaces. V1–V6 agreement on the 600-run: $\phi = 0.59$ — measured steerability is partly evaluation-relative; the main outcome is their union (cued $\vee$ uncued).

– Zero-shot judge (V6Z) rejected. Removing

1135	the judge’s three in-context examples lowers or ties every class (40/90 → 34/90 on v2b) — the examples calibrate rather than anchor.	
1136		
1137		
1138		
1139	? Graded judge (V8), secondary. Binary judging flattens sub-threshold movement (ragtime → lyre/melody; candlepower → binoculars): V8 re-judges the same generations on a 0/0.25/0.5/0.75/1 proximity rubric; r_{pb} with V6 = 0.81. Outcome-robustness: concreteness (−) and spectral concentration survive the outcome switch; Zipf shrinks to marginal — part of the binary-judge frequency effect is plausibly nameability. Main text keeps binary V1UV6 (closest to the RFM benchmark’s evaluation model). evaluation_prompts/wordnet_eval_v8.txt.	
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1153	F Appendix F: Choosing the judge model	
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1155	+ Judge shootout on a fixed 360-item set (pilot all-layer generations, v1+v4): local gpt-oss:20b 26/45 (v1), pod gpt-oss:20b 27/45, Claude Sonnet 5 26/45 — with identical per-class breakdowns for local-oss and Sonnet (10/15/1). Reproducible across machines; agrees with an independent frontier model.	
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1163	− Claude Haiku 4.5 is a systematically strict outlier (13/45; moods 15→5): of 49 disagreements with Sonnet, 48 are one-directional (under-crediting; it enforces fluency as a hard criterion). Rejected — and a demonstration that judge choice can move absolute rates ~2× while the class structure survives.	
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1171	+ Validation numbers (gpt-oss vs. Sonnet, $n=360$): agreement 86.7%/90.6% (v1/v4), Cohen’s $\kappa = 0.72/0.78$. Artifacts: oss/sonnet/haiku_judgments*.json.	
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1175	? Planned for the confirmatory study: re-validation on ~200 WordNet-distribution items; human spot-check of the ~25 gpt-oss/Sonnet disagreements.	
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1177		
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1179	+ Practical: gpt-oss:20b runs on the experiment GPU (ollama, ~1.5s/judgment) — judging costs GPU-hours, not API credits.	
1180		
1181		
	G Appendix G: Verbatim prompts	1182
	Extraction contrast (v4 frame). Each of the 100 general statements s appears once in each class. Positive class:	1183
		1184
		1185
	Imagine/assume: {concept}. What are your thoughts on the following statement? \n Statement: {s}	1186
		1187
		1188
	Negative class (bare):	1189
	What are your thoughts on the following statement? \n Statement: {s}	1190
		1191
	Both are wrapped in the model’s chat template as a single user turn.	1192
		1193
	Generation (evaluation) prompts.	1194
	V1: What is your favorite subject? Answer in 20 words or less.	1195
	V6: What would you like to talk about today? Answer in 20 words or less.	1196
		1197
		1198
	Greedy decoding, 50 new tokens, one generation per steering coefficient.	1199
		1200
	Judge template (V1; V6 differs only in the embedded question and its three examples’ responses). Model: gpt-oss:20b, temperature 0.	1201
		1202
		1203
	You are evaluating whether a language model has been successfully steered toward a target concept. You will be given:	1204
	1. An intended target concept (it may be of any kind: a feeling, a place, a profession, an object, an idea, etc.).	1205
	2. A response from the steered model to the question: "What is your favorite subject? Answer in 20 words or less."	1206
	Your task is to assess whether the response expresses the target concept.	1207
	Evaluation criteria:	1208
	1. The response counts as expressing the concept if it clearly centers on it, describes it, names it, OR manifestly embodies it (e.g., an emotional concept expressed through tone).	1209
	2. Generic content that would fit almost any concept of that kind does NOT count (e.g., naming the category but nothing specific to this concept).	1210
	3. Disfluent or fragmented wording does not disqualify a response if the concept is nevertheless clearly expressed.	1211
	4. The response does not need to name the concept literally if it is unmistakably about or in the manner of it.	1212
	Output format:	1213
	1. Score (0-1): 1 indicates successful steering and 0 indicates unsuccessful steering.	1214
	2. Explanation: A concise justification for the score.	1215
	Here are three examples:	1216
	1. Intended concept: volcano	1217
	Steered model response: "Volcanology, definitely! Eruptions, lava flows, the raw power beneath the crust."	1218
	Evaluation Output:	1219
	Score: 1	1220
	Explanation: The response centers unmistakably on volcanoes and their properties.	1221
	2. Intended concept: bitter	1222
	Steered model response: "Doesn't matter. Nothing's worth studying. People always let you down anyway."	1223
	Evaluation Output:	1224
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1248	Score: 1	?	Two candidate axes (now Discussion future	1298
1249	Explanation: The response does not name bitter-		work): (i) abstraction-depth — first layer	1299
1250	ness but clearly embodies a bitter tone.		that can steer the concept; (ii) composi-	1300
1251	3. Intended concept: Edinburgh		tional basis size (λ_1/trace). Between-class	1301
1252	Steered model response: "Geography! There are so		confounded (moods both more composite	1302
1253	many beautiful cities in the world to learn about."		and earlier-steering than places); indepen-	1303
1254	Evaluation Output:		dence is a within-supersense question.	1304
1255	Score: 0			
1256	Explanation: Generic geography content; nothing		– Extraction geometry cannot proxy	1305
1257	specific to Edinburgh.		first-steerable-depth: rotation and	1306
1258	Here is the concept: {concept} and here is the re-		alignment-to-deep proxies are concept-	1307
1259	sponse: {response}		invariant ($\sim 17\text{--}20$) and fail the steering	1308
			ground truth (cross-layer orthogonality, $\cos \approx 0.05$). A write property;	1309
1260	H Appendix Z1 (parked): Localizing the		measure by steering. nb_axes.ipynb;	1310
1261	concept — "where is Paris?"		plots/axes_independence.png.	1311
1262	(diagnostics)			1312
1263	+ ROME causal trace ("Paris is the capital of" $\rightarrow$ France): the fact lives at		?	1313
1264	the subject token in early-mid layers		Superposition vs. composition: a "combi-	1314
1265	($\sim 0\text{--}16$) and is relayed to the final		natorial basis $\times$ layer-specific sets" account	1315
1266	token only late ($\sim 14\text{--}31$). ROME-		is not ruled out; the SAE feature-count argu-	1316
1267	PARIS.png; diag_paris_rome.ipynb;		ment is the crux — open, not settled.	
1268	diag_paris_rome_ha.ipynb.		?	1317
1269			Stages of inference / lens framing (Lad-	1318
1270	+ This explains the RFM mismatch: RFM		Gurnee-Tegmark) as the principled naming	1319
1271	reads the last token, where subject-specific		of early/mid/late bands.	
1272	content arrives only late $\Rightarrow$ early extraction		?	1320
1273	captures an abstraction, late the precise in-		Not-all-linear (Engels et al.,	1321
1274	stance.		arXiv:2405.14860): some features are	1322
1275	+ SAE features confirm the model holds		circular/multi-dimensional — "which con-	1323
1276	Paris: clean Paris / France / city		cepts are globally-linear directions" is the	1324
1277	features at the steered layers (Good-		deepest version of the steerability ques-	1325
1278	fire l19; Llama-Scope L9/17/19, Neu-		tion; steering success is itself per-concept	1326
1279	ronpedia labels). diag_paris_sae.ipynb,		evidence for the global-linear reading.	
1280	diag_paris_sae_2.ipynb. The steering fail-			
1281	ure is extraction/geometry, not absence.		J Appendix Z3 (parked): The v_{spec}	1327
1282	?		probe — category-mean subtraction	1328
1283	Tegmark spatial probe (lat/lon linear		(R4, averaging variant)	1329
1284	probe): coordinates linearly decodable but			
1285	under-powered at $N=90$ cities (median		– Design and outcome: no rescue effect.	1330
1286	~ 2200 km). diag_paris_tegmark.ipynb.		$v_{\text{spec}}(c) = v(c) - \bar{v}_{\text{same}}$ supersense (leave-one-	1331
1287	+ Where "late" begins: logit-lens KL-		out), computed from the saved v2b di-	1332
1288	to-output gives stage boundaries to		rections with no re-extraction; per layer,	1333
1289	avoid the residual-sharpening tail.		unit-normalize the $K=1$ direction, sign-	1334
	diag_stages_late.ipynb.		align within supersense (saved eigenvec-	1335
			tor signs are arbitrary — 4–28% of	1336
1290	I Appendix Z2 (parked): Conceptual		within-category pairs have negative co-	1337
1291	threads		sine at mid layers; alignment via the	1338
1292	?		top eigenvector of the 25×25 Gram ma-	1339
1293	Abstract-early / precise-late hierarchy: the		trix), subtract the mean in the concept's	1340
1294	extraction layer sets the abstraction level;		own orientation, renormalize (dose un-	1341
1295	the precise concept lives late (weak steer-		changed). Six supersenses $\times 25$ (feeling,	1342
1296	ing) while the strong injection point is mid		food, person, location, artifact, body —	1343
1297	(concept abstract) — no single layer gives		NB this set swaps cognition out of the pre-	1344
	both.		specified featured six for body, the hard-	1345
			est cell; flag if the featured-cell definition	1346

is used elsewhere), all-layer default config, V1UV6; tag Lall_K1ev_def_fv2b_r4spec, builder build_r4spec_directions.py, RUN-LOG 013. Result: 25/150 steer vs. 30/150 baseline; +9 rescued / -14 lost / 16 retained (vs. ~ 5 expected under independence); McNemar $p \approx .4$. Per-supersense cell movements (feeling 9 $\rightarrow$ 6, body 2 $\rightarrow$ 3, ...) are within noise at $n=25$ /cell and are not interpreted.

– Reading: shared-component dominance is not the prevailing failure mode. The raw directions are $\sim 85\%$ category-shared (mean $|\cos|$ to the supersense mean .81–.90 across layers), yet removing that component flips only 9/120 failures. Masked concept-specific signal exists (beard: baseline collapses into excited stammering, v_{spec} at $c=0.8$ yields on-topic beard-styles content; similarly hay, fabric, residence) but is the exception.

? Side-yield (inspection-level): v_{spec} doubles as a frame-leak control. Baseline all-layer outputs carry the v2b register (“*excitedly* ...fascinated ...the wonders of”) at all coefficients; v_{spec} outputs do not — confirming the frame component lives in the category mean. Several lost feeling-class items (exuberance, sulk, mood) had baseline “successes” whose outputs never mention the concept but express frame-injected enthusiasm, which the symmetric V6 judge legitimately credits as embodiment; under v_{spec} these evaporate into neutral assistant-speak. Observation from reading generations, not a tested effect — an affect-blinded judge or larger feeling sample would be needed to quantify how much of feeling’s top R1 rank is frame-carried.

? Full variant (not run): within-category contrast re-extraction. Positives “loves Edinburgh”, negatives “loves {other city}” (same frame and statements both sides) — frame and parent cancel at the source; the only label-varying signal is the city. Steer single-layer (magn-calibrated, in-window) on ~ 5 cities: cities named $\Rightarrow$ the failure was extraction addressing; still generic $\Rightarrow$ instance content is not single-site injectable (assembly account).

? Re-running v_{spec} on the v4 directions was considered and decided against (jul10); the residual frame-leak risk for the v4 headline rates is instead acknowledged in Limitations.